

Task 1 — Lab Setup Report

Foundation & Environment Setup (Days 1–12)

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Task	Task 1 — Foundation & Environment Setup

1. Objective

The goal of this task was to build strong cybersecurity fundamentals — networking, Linux, and cryptography basics — and to set up a professional, isolated penetration testing lab using VirtualBox. The lab consists of a Kali Linux attacker machine and a Metasploitable2 target machine, connected on a private Host-only network so that all testing stays fully isolated from the real internet.

2. Lab Environment Overview

- Hypervisor: Oracle VirtualBox 7.2.14
- Attacker machine: Kali Linux 2026.2 (2048 MB RAM, 2 vCPUs)
- Target machine: Metasploitable2 (1024 MB RAM, deliberately vulnerable Ubuntu-based VM)
- Networking: Both VMs configured on a VirtualBox Host-only Adapter, isolating the lab from the host's real network and the internet
- Verified connectivity between attacker and target using ICMP (ping), and confirmed traffic visibility using Wireshark

3. Kali Linux — Attacker Machine

Kali Linux was imported into VirtualBox and configured with a Host-only network adapter. The screenshot below confirms the VM is running and the desktop environment has loaded successfully.

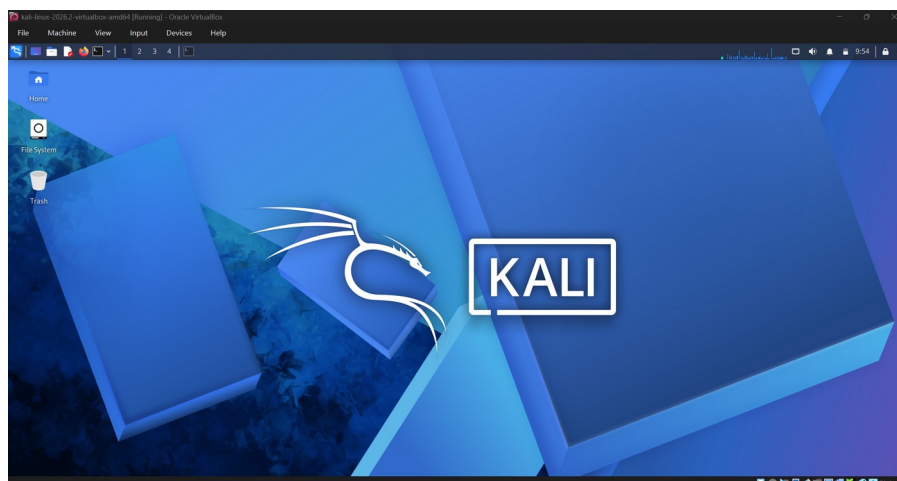
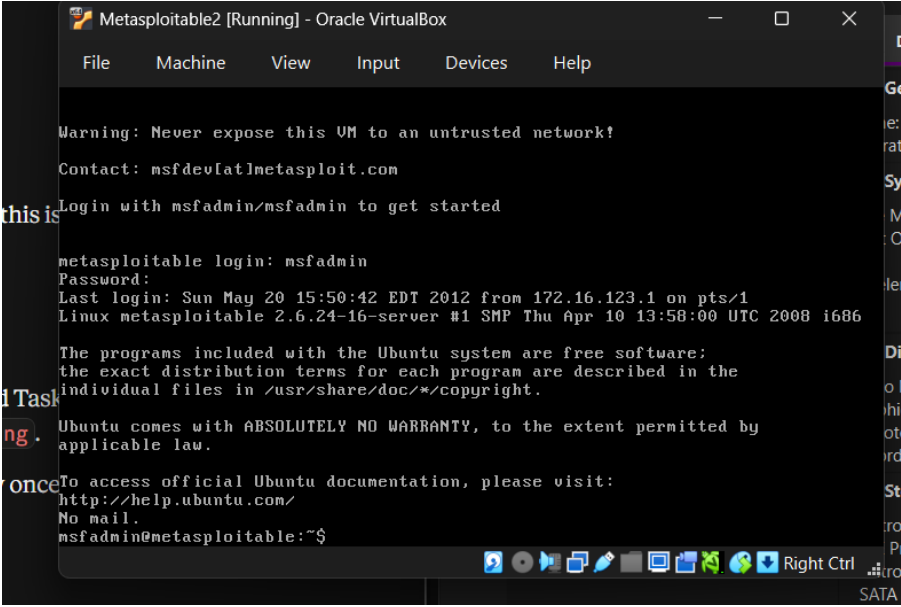


Figure 1: Kali Linux desktop running inside VirtualBox, confirming a successful VM import and boot.

4. Metasploitable2 — Target Machine

Metasploitable2 was added to VirtualBox using its existing .vmdk disk image and placed on the same Host-only network as Kali. It was then booted and logged into using the default credentials (msfadmin / msfadmin), confirming the target machine is reachable and operational.



```
Metasploitable2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

metasploitable login: msfadmin
Password:
Last login: Sun May 20 15:50:42 EDT 2012 from 172.16.123.1 on pts/1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

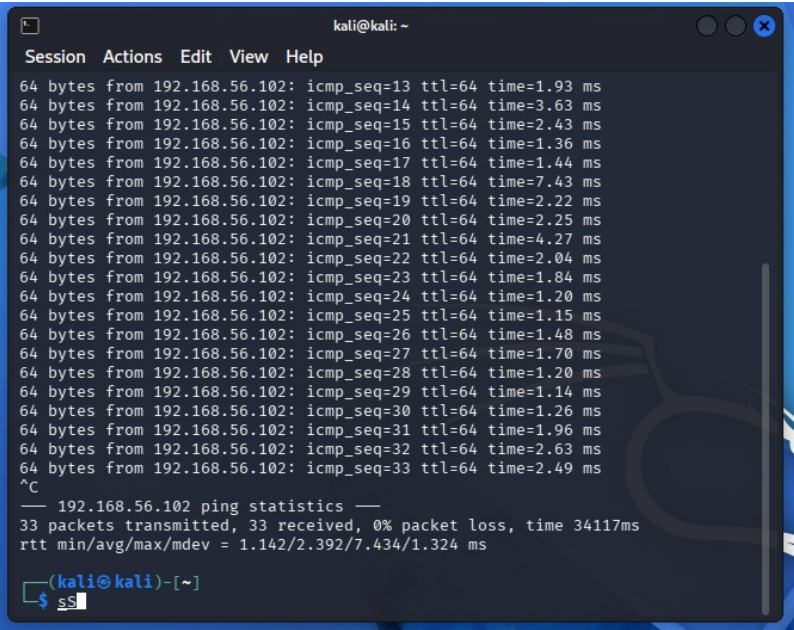
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$
```

Figure 2: Successful login to the Metasploitable2 target VM.

5. Network Connectivity Test

With both VMs configured on the same Host-only Adapter, connectivity was verified from Kali by pinging Metasploitable2's IP address (192.168.56.102). The result below shows 33 packets transmitted with 0% packet loss, confirming the isolated lab network is functioning correctly.



```
kali@kali: ~
Session Actions Edit View Help

64 bytes from 192.168.56.102: icmp_seq=13 ttl=64 time=1.93 ms
64 bytes from 192.168.56.102: icmp_seq=14 ttl=64 time=3.63 ms
64 bytes from 192.168.56.102: icmp_seq=15 ttl=64 time=2.43 ms
64 bytes from 192.168.56.102: icmp_seq=16 ttl=64 time=1.36 ms
64 bytes from 192.168.56.102: icmp_seq=17 ttl=64 time=1.44 ms
64 bytes from 192.168.56.102: icmp_seq=18 ttl=64 time=7.43 ms
64 bytes from 192.168.56.102: icmp_seq=19 ttl=64 time=2.22 ms
64 bytes from 192.168.56.102: icmp_seq=20 ttl=64 time=2.25 ms
64 bytes from 192.168.56.102: icmp_seq=21 ttl=64 time=4.27 ms
64 bytes from 192.168.56.102: icmp_seq=22 ttl=64 time=2.04 ms
64 bytes from 192.168.56.102: icmp_seq=23 ttl=64 time=1.84 ms
64 bytes from 192.168.56.102: icmp_seq=24 ttl=64 time=1.20 ms
64 bytes from 192.168.56.102: icmp_seq=25 ttl=64 time=1.15 ms
64 bytes from 192.168.56.102: icmp_seq=26 ttl=64 time=1.48 ms
64 bytes from 192.168.56.102: icmp_seq=27 ttl=64 time=1.70 ms
64 bytes from 192.168.56.102: icmp_seq=28 ttl=64 time=1.20 ms
64 bytes from 192.168.56.102: icmp_seq=29 ttl=64 time=1.14 ms
64 bytes from 192.168.56.102: icmp_seq=30 ttl=64 time=1.26 ms
64 bytes from 192.168.56.102: icmp_seq=31 ttl=64 time=1.96 ms
64 bytes from 192.168.56.102: icmp_seq=32 ttl=64 time=2.63 ms
64 bytes from 192.168.56.102: icmp_seq=33 ttl=64 time=2.49 ms
^C
— 192.168.56.102 ping statistics —
33 packets transmitted, 33 received, 0% packet loss, time 34117ms
rtt min/avg/max/mdev = 1.142/2.392/7.434/1.324 ms

(kali@kali)-[~]
$ ss
```

Figure 3: Successful ping test from Kali Linux to Metasploitable2 (0% packet loss).

6. Packet Analysis with Wireshark

Wireshark was used on Kali to capture live traffic on the eth0 interface while the ping test was running. Applying an icmp display filter isolates the ping request/reply packets, and the packet detail pane confirms the Ethernet, IP, and ICMP protocol layers involved in the exchange.

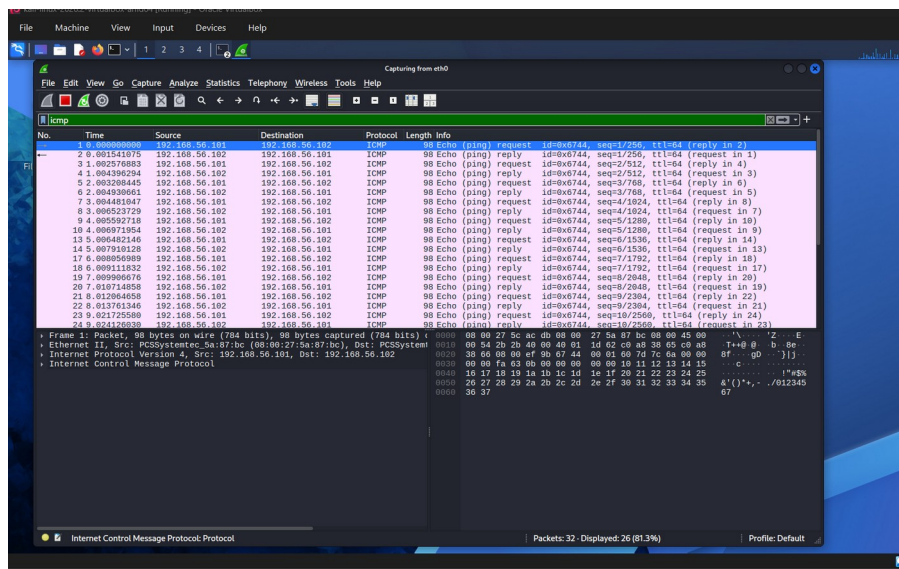


Figure 4: Wireshark capture on eth0, filtered to ICMP traffic, showing ping request/reply pairs between 192.168.56.101 (Kali) and 192.168.56.102 (Metasploitable2).

7. Tool Familiarization — Nmap & Netcat

As part of getting familiar with the core toolset ahead of Task 2, Nmap and Netcat were run against the Metasploitable2 target. An Nmap service/version scan (nmap -sV -Pn) identified a large number of open ports and outdated service versions — including FTP, SSH, Telnet, SMTP, HTTP (Apache 2.2.8), Samba, MySQL, PostgreSQL, and VNC — confirming why Metasploitable2 is a suitable target for practicing vulnerability identification. Netcat was then used to manually verify that port 80 (HTTP) was open and reachable, confirming the TCP connection succeeded.

- Cryptography basics: symmetric vs asymmetric encryption, hashing, and SSL/TLS certificates
- Tool familiarization: VirtualBox, Kali Linux, Wireshark, and the Host-only networking model used for safe, isolated lab testing

10. Conclusion

Task 1 is complete. A fully isolated penetration testing lab has been built with a working Kali Linux attacker VM and a Metasploitable2 target VM, both verified to communicate over a private Host-only network. Core tools — Wireshark, Nmap, Netcat, and Burp Suite — have all been installed and tested successfully. This environment forms the foundation for all subsequent tasks, including network scanning, web application security testing, exploitation, and the final capstone project.